In [5]:
```
import os
os.environ["OMP_NUM_THREADS"] = "1"
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans
```

In [6]:
```
df = pd.read_csv('Mall_Customers.csv')
df.head()
```

Out[6]:

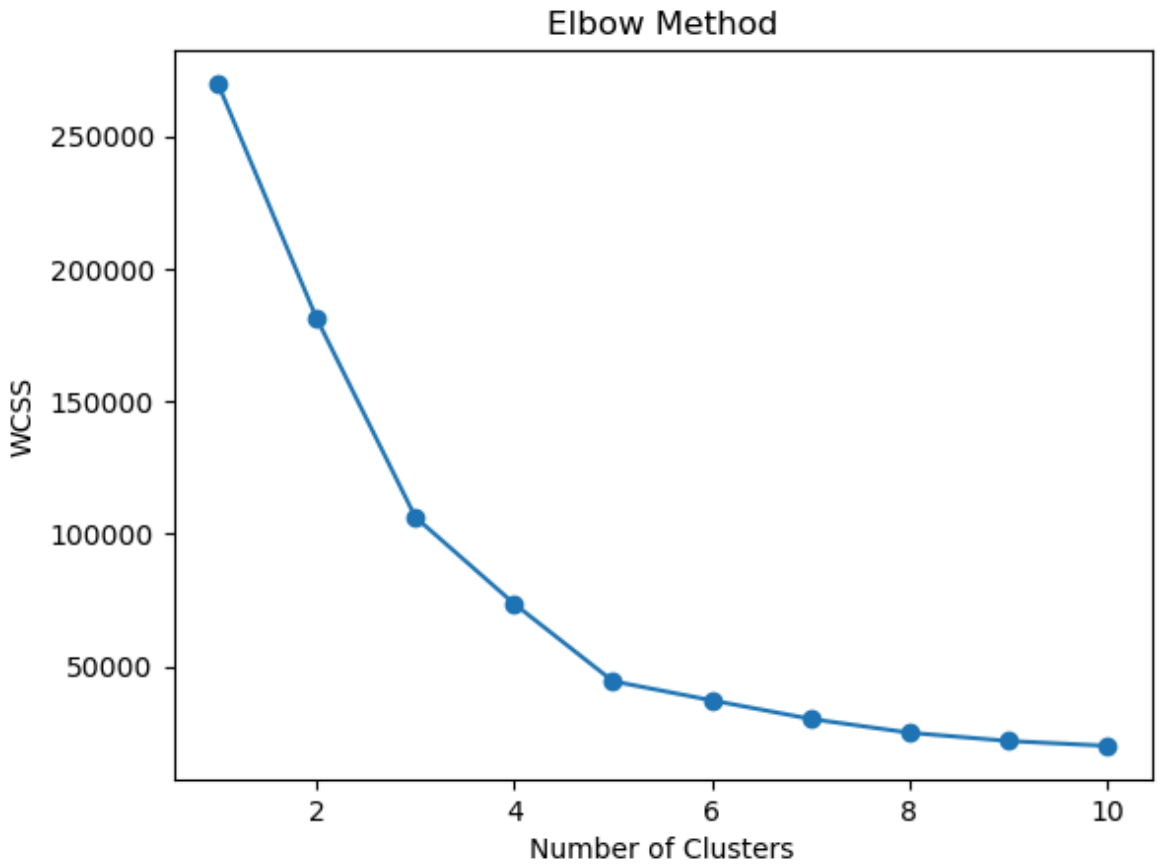
	CustomerID	Gender	Age	Annual Income (k$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40

In [8]:
```
X = df[['Annual Income (k$)','Spending Score (1-100)']]
```

In [13]:
```
# Elbow Method Find the optimal K
# WCSS stands for within-cluster sum of squares -It measures how far data
wcss = []

for k in range(1, 11):
    model = KMeans(n_clusters=k, random_state=42, n_init=10)
    model.fit(X)
    wcss.append(model.inertia_)
```

In [15]:
```
plt.plot(range(1, 11), wcss, marker='o')
plt.xlabel('Number of Clusters')
plt.ylabel('WCSS')
plt.title('Elbow Method')
plt.show()
```



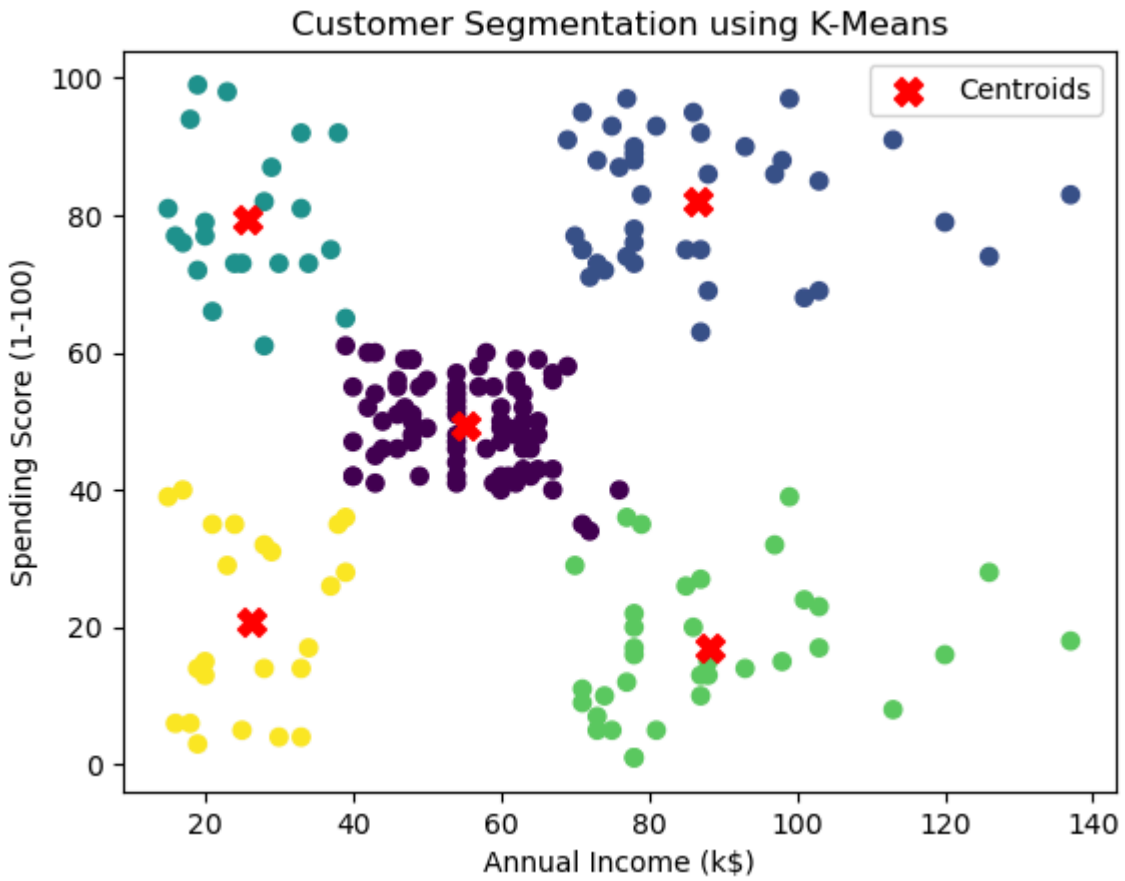
In [18]:
```
#step 6
model = KMeans(n_clusters=5, random_state=42, n_init=10)

df['Cluster'] = model.fit_predict(X)
```

In [19]:
```
# Step 7: Visualize clusters and centroids
plt.scatter(
    X['Annual Income (k$)'],
    X['Spending Score (1-100)'],
    c=df['Cluster']
)

# Plot the centroids
plt.scatter(
    model.cluster_centers_[:, 0],
    model.cluster_centers_[:, 1],
    color='red',
    marker='X',
    s=100,
    label='Centroids'
)

plt.xlabel('Annual Income (k$)')
plt.ylabel('Spending Score (1-100)')
plt.title('Customer Segmentation using K-Means')
plt.legend()
plt.show()
```



In [20]:
```
df
```

Out[20]:

	CustomerID	Gender	Age	Annual Income (k$)	Spending Score (1-100)	Cluster
0	1	Male	19	15	39	4
1	2	Male	21	15	81	2
2	3	Female	20	16	6	4
3	4	Female	23	16	77	2
4	5	Female	31	17	40	4
...	...	...	...	...	...	...
195	196	Female	35	120	79	1
196	197	Female	45	126	28	3
197	198	Male	32	126	74	1
198	199	Male	32	137	18	3
199	200	Male	30	137	83	1

200 rows × 6 columns

In [ ]:
```
```